

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K}$.

8.1. For the two matrices

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

determine the axis of rotation and the rotation angle.

8.2. Show that the transformation

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \mapsto \frac{1}{45} \begin{bmatrix} 37 & -20 & -16 & 15 \\ 16 & 40 & -13 & -30 \\ 20 & 5 & 40 & 30 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

is a glide-rotation. Find the rotation angle and the rotation axis.

8.3. Show that the transformation

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \mapsto \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 & 4 \\ -2 & -1 & 2 & 2 \\ 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

is a glide-reflection. Find the reflecting plane and the translation vector.

8.4. In dimension 3, show that the composition of a rotation with 60° around the x -axis followed by a rotation with 30° in the z -axis is a rotation. Moreover, determine the rotation angle.

8.5. Using rotations around the coordinate axes, give a parametrization of a cylinder with axis $\mathbb{R}(0, 1, 1)$ and diameter $\sqrt{2}$.

6.1. For the two matrices

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \frac{1}{11} \begin{bmatrix} -9 & -2 & 6 \\ 6 & -6 & 7 \\ 2 & 9 & 6 \end{bmatrix}$$

determine the axis of rotation and the rotation angle.

- the axis of rotation is the line passing through the origin in the direction of the eigenvectors for the eigenvalue 1

this is obtained by solving $A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \Leftrightarrow \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & -2 & -1 \\ -2 & 1 & 2 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\Leftrightarrow \begin{bmatrix} -4 & 2 & -2 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{bmatrix} -4 & 2 & -2 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 1 \\ -2 & -5 & -1 \\ -2 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 1 \\ 0 & -6 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow y=0 \quad \begin{cases} 2x + z = 0 \\ z = -2x \end{cases}$$

$\Rightarrow$ eigenspace for $\lambda=1$ is $V_1 = \{ (t, 0, -2t) : t \in \mathbb{R} \}$ this is a line passing through the origin, it is the rotation axis
 the eigenvectors are the non-zero vectors in V_1

- the angle of rotation θ is determined by

$$\text{tr}(A) = 1 + 2 \cos \theta \quad (\Rightarrow) \quad \cos \theta = \frac{\text{tr}(A) - 1}{2} = \frac{-\frac{1}{3} - 1}{2} = -\frac{2}{3}$$

$$\text{so} \quad \theta = \arccos\left(-\frac{2}{3}\right)$$

the calculation for B is similar.

6.2. Show that the transformation

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \mapsto \frac{1}{45} \begin{bmatrix} 37 & -20 & -16 & 15 \\ 16 & 40 & -13 & -30 \\ 20 & 5 & 40 & 30 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

is a glide-rotation. Find the rotation angle and the rotation axis.

Let $A = \frac{1}{45} \begin{bmatrix} 37 & -20 & -16 \\ 16 & 40 & -13 \\ 20 & 5 & 40 \end{bmatrix}$ be the matrix of the linear part

- Since $A \cdot A^T = I_3$ and $\det A = 1$ the matrix is in $SO(3)$.
The affine map is therefore a direct isometry.
- Since $A \neq I_3$, the map is not the identity or a translation.
- By the classification of direct isometries it can only be a rotation or a glide rotation.
In the first case it fixes the rotation axis.
In the second case it doesn't fix any point.
- We investigate the fixed points of the map.

$$\frac{1}{45} \begin{bmatrix} 37 & -20 & -16 \\ 16 & 40 & -13 \\ 20 & 5 & 40 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{45} \begin{bmatrix} 15 \\ -30 \\ 30 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} -5 & -20 & -16 \\ 16 & -5 & -13 \\ 20 & 5 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} 15 \\ -30 \\ 30 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We know that the rank of the matrix of coefficients is 2, since A is a rotation (but you can check this directly)

- So, it suffices to check the rank of the extended matrix
 $\rightarrow$ this is either $\begin{cases} 2 \\ \text{or} \\ 3 \end{cases}$ in which case we have a rotation (since there are fixed points)
in which case we have a glide rotation (since there are no fixed points)

$\rightarrow$ a calculation shows that we are in the second case

- Alternatively, one can calculate the rotation axis for the rotation represented by A and check if the translation part of the map $\frac{1}{45} \begin{bmatrix} 15 \\ -20 \\ 30 \end{bmatrix}$ is orthogonal to the axis.

$\begin{cases} \rightarrow \text{if it is, then we have a rotation} \\ \rightarrow \text{if it is not, then we have a glide-rotation} \end{cases}$

6.3. Show that the transformation

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \mapsto \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 & 4 \\ -2 & -1 & 2 & 2 \\ 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

is a glide-reflection. Find the reflecting plane and the translation vector.

Let $A = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ -2 & -1 & 2 \\ 1 & 2 & 2 \end{bmatrix}$ be the matrix of the linear part of the map.

- Since $A \cdot A^T = I_3$ we have $A \in O(3)$
- In addition $\det A = -1$ so the map is an indirect isometry.
- By the classification of indirect isometries we have

either $\begin{cases} \text{reflection} \\ \text{glide reflection} \\ \text{rotation-reflection} \end{cases}$

we need to figure out which it is

• Our map is $\phi(P) = AP + v$ where $v = \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

• We first investigate the linear part $\psi(P) = AP$

Since $A \in O(3) \setminus SO(3)$ it admits -1 as an eigenvalue and the eigenvectors for -1 are normal vectors for planes which are invariant under ψ

• we determine the eigenvectors for -1

$$Av = -v \Leftrightarrow \frac{1}{3} \begin{bmatrix} 5 & -2 & 1 \\ -2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -2 & 1 \\ -2 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 5 \\ 0 & -12 & -24 \\ 0 & 6 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x+z=0 \\ y+2z=0 \end{cases}$$

$\Rightarrow$ the eigenvectors are spanned by $w = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

• the other two eigenvalues for ψ are

either $\begin{cases} \text{both } 1 \end{cases}$ in which case ψ is a reflection
or $\begin{cases} \text{both complex} \end{cases}$ in which case ψ is a rotation

• to find the eigenvalues, we calculate the characteristic polynomial

$$\det(A - \lambda I) = \det \frac{1}{3} \begin{bmatrix} 2-3\lambda & -2 & 1 \\ -2 & -1-3\lambda & 2 \\ 1 & 2 & 2-3\lambda \end{bmatrix} = -(\lambda-1)^2(\lambda+1)$$

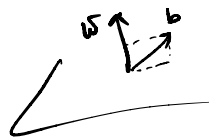
so it admits 1 as an eigenvalue $\Rightarrow \psi$ is a reflection

• returning to ϕ , which is the reflection ψ followed by the translation T_v where $v = \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

• by the results in the lecture, the reflection plane is

$$\pi : \langle w, x \rangle = \frac{1}{2} \langle w, v \rangle$$

$$\Leftrightarrow \pi : x + 2y - z = 1$$



- moreover, the translation vector is the component of b which is parallel to π , i.e.

$$v - \text{Pr}_w^\perp(v) = v - \frac{\langle w, v \rangle}{\langle w, w \rangle} w = \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{6} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

6.4. In dimension 3, show that the composition of a rotation with 60° around the x -axis followed by a rotation with 30° in the z -axis is a rotation. Moreover, determine the rotation angle.

Rotation with 60° around Ox : $R_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$

Rotation with 30° around Oz : $R_2 = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$

R_1 and R_2 have determinant 1

$$\Rightarrow \det(R_2 R_1) = 1$$

$\Rightarrow$ the composition is a direct isometry

Both rotations fix the origin $\Rightarrow$ so does the composition
Then, since it fixes a point, by the classification of direct isometries it has to be a rotation

To find the rotation angle we use the trace-formula

$$R = R_2 R_1 = \begin{bmatrix} \frac{\sqrt{3}}{2} & -1/4 & \frac{\sqrt{3}}{4} \\ 1/2 & \frac{\sqrt{3}}{4} & -3/4 \\ 0 & \frac{\sqrt{3}}{2} & 1/2 \end{bmatrix}$$

$$\Rightarrow \text{Tr}(R) = \frac{3\sqrt{3} + 2}{4}$$

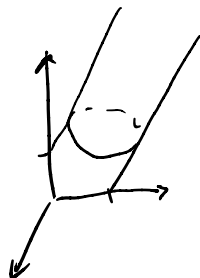
$$\Rightarrow \cos(\theta) = \frac{\text{Tr}(R) - 1}{2} = \frac{3\sqrt{3} - 2}{8} \quad (\leadsto \theta \approx 66.45^\circ)$$

6.5. Using rotations around the coordinate axes, give a parametrization of a cylinder with axis $\mathbb{R}(0,1,1)$ and diameter $\sqrt{2}$.

The cylinder contains the line

$$l: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

and has axis $l': \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$



We may construct a cylinder $\tilde{\mathcal{C}}$ of diam. $\sqrt{2}$ and axis Oz and then rotate it in the right position

$$\tilde{\mathcal{C}} \ni \tilde{l}: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ \sqrt{2}/2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\tilde{\mathcal{C}} = \text{Rot}_{k,0}(\tilde{l}) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \sqrt{2}/2 \\ t \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} \sin\theta \\ \frac{\sqrt{2}}{2} \cos\theta \\ t \end{bmatrix}$$

$$\Rightarrow \mathcal{C} = \text{Rot}_{i,45^\circ}(\tilde{\mathcal{C}}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2}/2 & \sqrt{2}/2 \\ 0 & -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} -\frac{\sqrt{2}}{2} \sin\theta \\ \frac{\sqrt{2}}{2} \cos\theta \\ t \end{bmatrix} = \dots$$